

Sprint Plan & Team Charter

Retail Customer Intelligence Platform — a 4-week delivery of a Mini Enterprise Data Platform: Dashboard + RFM + Churn Prediction + CI/CD, built SDLC-first.

5 members

4 sprints

Airflow · dbt · Docker

MLflow · Streamlit

GitHub Actions

Module 01 · Project #22 + #20 · Prepared 02 June 2026

0 Sprint Goal & Guiding Principle

Goal: In 4 weeks, demo a Mini Enterprise Data Platform with dashboard + RFM + churn prediction + basic CI/CD. We do **not** attempt the full P0 → P3 roadmap; we deliver a credible vertical slice.

Guiding principle — SDLC over model. Many teams jump straight into training XGBoost on day one. For this project, what makes a reviewer call it "enterprise" is the chain **BRD** → **UML** → **Star Schema** → **Airflow** → **dbt** → **Testing** → **CI/CD**, not the churn model. The model is only ~15–20% of total project value.

1 Team & Roles

TECH LEAD

Phan Văn Tiến

BRD · UML · architecture · sprint planning · MLflow · business impact

AI ENG · DATA

Võ Ngọc Gia Bảo

Star schema · dbt models · feature mart · KPI marts · dashboard

AI ENG · MODEL

Phúc Nhân Nguyễn

RFM logic · feature engineering · churn model · explainability

AI ENG · PIPELINE

Ngọc Phương

Repo · Docker Compose · Airflow DAGs · CI/CD

QA · REVIEWER

Hoàng Đức Kiên

DoD · test plan · data/model/UAT validation · final review

2 Sprint Timeline

Week 1 — Discovery & Design · lay the SDLC foundation

Owner	Tasks	Deliverables
LEAD	BRD · scope · backlog · sprint planning · review architecture	<code>docs/BRD.md</code> , <code>Solution_Architecture.md</code> , <code>Project_Plan.md</code>
DATA	Data profiling · data dictionary · source analysis · quality rules	<code>docs/data_dictionary.xlsx</code> , <code>notebooks/eda.ipynb</code>
PIPELINE	Repo setup · Docker Compose · Airflow skeleton · folder structure	<code>docker-compose.yml</code> , <code>airflow/</code> , <code>dbt/</code> , <code>data/</code>
MODEL	Churn definition · feature brainstorming · RFM logic design	<code>docs/ml_design.md</code>
QA	Definition of Done · test checklist · review BRD	<code>docs/test_plan.md</code>

Week 2 — Data Platform MVP · raw → star schema → RFM

Owner	Tasks	Output
DATA	Build Bronze → Silver → Gold; star schema tables	<code>dim_customer</code> , <code>dim_product</code> , <code>dim_date</code> , <code>dim_country</code> , <code>fact_transactions</code>
PIPELINE	Airflow DAG: ingest → clean → dbt run → dbt test	Working scheduled DAG
MODEL	Build RFM; segment labelling	<code>customer_rfm</code> (Champions / Loyal / At Risk / Lost)
QA	Validate row count · null · duplicate	Validation log

End of Week 2 Demo:

CSV → Airflow → dbt → Star Schema → RFM Table

Week 3 — ML Layer · features → model → churn probability

Owner	Tasks	Output
MODEL	Feature engineering (Recency, Frequency, Monetary, AOV, LTV); train LR + XGBoost	Trained models + metrics
DATA	Build feature mart	<code>mart_customer_features</code>
PIPELINE	Add DAG tasks: <code>train_model</code> , <code>score_customers</code>	Extended DAG
LEAD	MLflow integration (experiments + registry)	MLflow tracking server
QA	Model tests: schema · range · AUC threshold	Model test report

End of Week 3 Demo:

Customer → Features → Model → Churn Probability

Week 4 — Dashboard + CI/CD + Final Demo · ship it

Owner	Tasks	Output
DATA	KPI marts: revenue · retention · cohort; build dashboard	Streamlit dashboard
MODEL	Explainability — top churn drivers	Feature importance view
PIPELINE	GitHub Actions: lint → dbt test → pytest → docker build	.github/workflows/ci.yml
QA	UAT: dashboard · data · model validation	UAT sign-off
LEAD	Business Impact Report (Champions / At Risk revenue, top 5% churn)	docs/business_impact.md

Final Demo:

Raw CSV → Airflow → dbt Star Schema → RFM → ML Churn → Dashboard → Retention List + Business Impact

3 Component Ownership Matrix

Component	Owner
BRD / UML	Tech Lead
Star Schema	Data Engineer
dbt Models	Data Engineer
Airflow	Pipeline Engineer
Docker	Pipeline Engineer
ML Model	Model Engineer
MLflow	Model Engineer + Lead
Dashboard	Data Engineer
Testing	QA
CI/CD	Pipeline Engineer
Final Review	QA + Lead

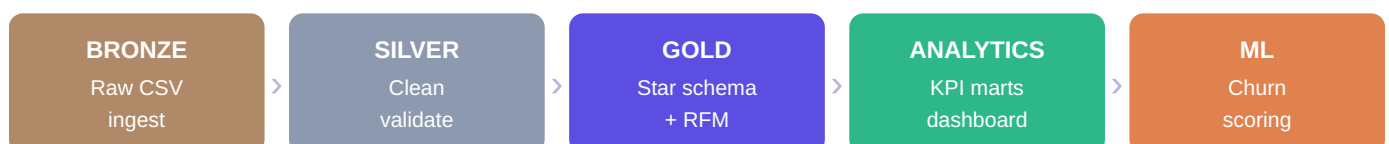
4 Build Priority — Path to 10/10

If forced to cut scope, build in this order. The first seven items are what earn the "enterprise" judgment.

- 1 **BRD + UML** — the business framing that frames everything
- 2 **Star Schema** — dimensional model, the Data Engineer money-shot
- 3 **Airflow** — orchestration backbone
- 4 **dbt** — transforms + tests as code
- 5 **Dashboard** — the visible deliverable stakeholders see
- 6 **Docker** — [docker-compose up](#) , reproducible for reviewers
- 7 **Churn Model** — predictive layer (only now)
- 8 **MLflow** — experiment tracking + registry
- 9 **GitHub Actions** — CI/CD automation
- 10 **Monitoring** — data + model observability (stretch)

Reminder: the enterprise feel comes from the chain BRD → UML → Star Schema → Airflow → dbt → Testing → CI/CD — *not* from the churn model. Resist the urge to over-invest in modelling early.

5 MVP Architecture (4-Week Build)



```
docker-compose up
```



```
CI: GitHub Actions -> lint · dbt test · pytest · docker build
```

Stack is fully local/containerized for fast iteration and reviewer reproducibility within the 4-week window — no cloud provisioning overhead.

6 Appendix — Production Cloud Path (Azure)

Out of scope for the 4-week build, documented to show cloud vision. The local stack maps cleanly onto Azure-native managed services (no Databricks).

Local (MVP)	Azure (production)	Note
Airflow (Docker)	Azure Data Factory	Managed orchestration / scheduling
Local volume / DuckDB	ADLS Gen2 (Bronze/Silver/Gold)	Lakehouse storage
dbt + train (container)	Container Apps Job (cron)	Scale-to-zero, pay-per-run
Postgres (Docker)	Azure DB for PostgreSQL	Serving marts (+ pgvector)
Streamlit (Docker)	Container Apps (App)	Serverless dashboard hosting
.env file	Azure Key Vault	Secrets + Managed Identity
local images	Azure Container Registry	Image registry
logs to console	Azure Monitor + Log Analytics	Data/model monitoring + alerts

Aligns with the team lead's AZ-104 / DP-750 certification track — the same architecture doubles as cloud-skills demonstration.